

Highlights

Community detection in signed networks: Fast label propagation algorithm and post-hoc statistical analysis

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- Fast label propagation algorithm for community detection in signed networks.
- Extremely fast detection with high partition quality.
- Outperforms existing algorithms across 30 empirical signed networks.
- Normalized partition quality metrics and post-hoc statistical analysis.
- Complete source code and results enable reproducible comparisons.

Community detection in signed networks: Fast label propagation algorithm and post-hoc statistical analysis

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Abstract

Community detection in signed networks seeks to identify a balanced partition of nodes, with positive edges within clusters and negative edges between them. Applications range from identifying polarized groups in political discussions to uncovering alliances in international relations. In this paper, we propose an extension of fast label propagation for community detection in signed networks. We evaluate the algorithm on synthetic signed graphs and 30 empirical signed networks. We show that it identifies higher-quality partitions than existing algorithms, as measured by the frustration and external–internal indices, while also being substantially faster. In addition, we introduce normalized variants of established partition quality metrics that balance the contributions of positive and negative edges in networks with highly unequal proportions of the two. We also present a rank-based post-hoc statistical analysis for comparing multiple algorithms across a large number of networks. Finally, to enable reproducible comparisons, we will make the complete source code and experimental results publicly available.

Keywords: signed networks, community detection, fast label propagation algorithm, post-hoc statistical analysis

1. Introduction

Networks or graphs are a popular tool for analyzing real-world systems across various scientific fields (Newman, 2018), ranging from friendship networks in sociology to collaboration networks in scientometrics. A network consists of entities represented by nodes and interactions between them represented by edges. Whereas standard networks represent interactions between entities without distinguishing their nature, signed networks distinguish between positive and negative interactions. Examples include friendship and hostility in social networks (Ruiz-García et al., 2023), alliances and rivalries in international relations (Doreian and Mrvar, 2015), mutualism and competition in ecological networks (Allesina and Tang, 2012), and reinforcing and conflicting beliefs in psychological networks (Aiyappa et al., 2024).

Seminal work on signed networks dates back to Heider (1946), who introduced structural balance theory, proposing that social relationships tend toward configurations in which the friend of a friend is a friend and the enemy of an enemy is also a friend. Cartwright and Harary (1956) later formalized structural balance theory in graph-theoretic terms, showing that a balanced signed network can be partitioned into two clusters with positive edges within clusters and negative edges between them. The latter is known as strong balance, while weak balance generalizes this condition to allow for any number of clusters (Davis, 1967). These foundational ideas laid the groundwork for the study of signed networks.

Community detection is a frequent task in network analysis, aimed at identifying groups of nodes that are densely connected within and sparsely connected between (Girvan and Newman, 2002). A wide variety of approaches for community detection have been proposed in the literature, reflecting different interpretations of what constitutes a community (Fortunato and Hric, 2016). In signed networks, community detection typically seeks a balanced partition, in which positive edges lie within clusters and negative edges between them. Identifying such partitions can reveal groups of nodes that tend to cooperate, agree, or form alliances internally while competing, disagreeing, or forming rivalries with other groups. Despite its broad applicability, considerably fewer algorithms have been developed for community detection in signed than in unsigned networks (Diaz-Diaz et al., 2025).

In this paper, we extend the fast label propagation algorithm of Traag and Šubelj (2023) to community detection in signed networks. We evaluate the proposed algorithm on synthetic graphs with planted community structure to test its detection ability, and on synthetic random graphs with no community structure to test its robustness. Furthermore, we compare the algorithm with existing ones on 30 empirical signed networks. The comparison includes a diverse set of algorithms for signed networks. Since empirical signed networks contain predominantly positive edges, we also consider popular algorithms for unsigned networks, which may still perform well in such settings.

The evaluation is based on standard partition quality metrics. However, due to the predominance of positive edges, we argue that these metrics may provide a less meaningful evaluation and

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introduce normalized variants that balance the contributions of positive and negative edges. We also consider the running time of the algorithms as an evaluation metric, which is particularly relevant given the large size of many empirical networks. Finally, since our comparison results in almost four hundred values per metric, direct comparison becomes cumbersome. To this end, we adopt methods from the machine learning literature (Demšar, 2006) that enable straightforward post-hoc statistical comparison of the algorithms.

The rest of the paper is structured as follows. Section 2 presents the methodology, including notation and quality metrics used for post-hoc analysis of the results. Section 3 presents a fast label propagation algorithm for community detection in signed networks. Section 4 evaluates the proposed algorithm on synthetic graphs and empirical networks, including comparisons with existing algorithms. Section 5 summarizes the main findings and their practical implications.

2. Methodology

2.1. Graph notations

Let a signed network be represented by an unweighted, undirected multigraph $G(V, E)$ with neither self-loops nor isolated nodes, where $V = \{1, \dots, n\}$ denotes the set of nodes and $m = |E|$ is the number of edges. Each edge is assigned either a positive or a negative sign. Let m^+ and m^- denote the numbers of positive and negative edges, respectively, such that $m = m^+ + m^-$. Note that, in an unsigned network, every edge is assumed to have a positive sign, and thus $m^- = 0$. Next, let $A \in \mathbb{Z}^{n \times n}$ be the adjacency matrix of the network, decomposed as $A = A^+ - A^-$, where A_{ij}^+ and A_{ij}^- denote the numbers of positive and negative edges between nodes $i, j \in V$, respectively. Since the network is represented by an undirected graph, $A = A^T$. The neighborhood of node $i \in V$ is denoted by $\Gamma_i = \{j : A_{ij}^+ > 0 \vee A_{ij}^- > 0\}$. Finally, let $k_i = k_i^+ + k_i^-$ be the degree of node i , where k_i^+ and k_i^- denote the numbers of positive and negative edges incident to node $i \in V$, respectively.

2.2. Partition metrics

Consider a partition of the nodes V into disjoint clusters, and let c_i denote the cluster label of node $i \in V$. Community detection in signed networks seeks to identify a balanced partition, in which positive edges lie within clusters and negative edges lie between clusters. To assess the quality of partitions produced by different community detection algorithms, we employ different metrics that measure their imbalance or frustration.

The frustration index $FI \in [0, 1]$ is rooted in the spin glass theory of statistical mechanics and measures the fraction of edges that must be removed or have their signs flipped to obtain a frustration-free partition (Harary, 1959). It is defined as

$$FI = \frac{1}{m} \sum_{i < j} A_{ij}^+ [1 - \delta(c_i, c_j)] + A_{ij}^- \delta(c_i, c_j), \quad (1)$$

where $\delta(\cdot, \cdot)$ is the Kronecker delta function. Note that lower values of FI correspond to more balanced partitions.

The external–internal index is a segregation metric that measures the degree of homophily or heterophily of a partition (Krackhardt and Stern, 1988). It can be extended to signed networks by considering positive and negative edges separately. The unnormalized external–internal index of positive edges EI^+ is defined as the difference between the numbers of inter-cluster and intra-cluster positive edges.

$$\begin{aligned} EI^+ &= \sum_{i < j} A_{ij}^+ [1 - \delta(c_i, c_j)] - A_{ij}^+ \delta(c_i, c_j) \\ &= m^+ - 2 \sum_{i < j} A_{ij}^+ \delta(c_i, c_j) \end{aligned}$$

Similarly, the unnormalized external–internal index of negative edges EI^- is defined as

$$EI^- = m^- - 2 \sum_{i < j} A_{ij}^- \delta(c_i, c_j).$$

Finally, the signed external–internal index $EI \in [-1, 1]$ favors intra-cluster positive edges and inter-cluster negative edges and is defined by

$$EI = \frac{EI^+ - EI^-}{m}, \quad (2)$$

where lower values indicate partitions with stronger segregation.

Modularity is a measure of the quality of community structure in unsigned networks (Newman and Girvan, 2004). It compares the observed number of intra-cluster edges with the expected number of such edges under a selected random graph model, where the configuration model preserving the same degree sequence is the standard choice (Newman et al., 2001). As with the external–internal index, modularity can be extended to signed networks by considering positive and negative edges separately (Gómez et al., 2009). The unnormalized modularity of positive edges Q^+ is defined as

$$Q^+ = \sum_{i < j} \left(A_{ij}^+ - \frac{k_i^+ k_j^+}{2m^+} \right) \delta(c_i, c_j)$$

and the unnormalized modularity of negative edges Q^- as

$$Q^- = \sum_{i < j} \left(A_{ij}^- - \frac{k_i^- k_j^-}{2m^-} \right) \delta(c_i, c_j).$$

Finally, the signed modularity $Q \in [-1, 1]$ favors a large positive component Q^+ and a small negative component Q^- and is defined by

$$Q = \frac{Q^+ - Q^-}{m}, \quad (3)$$

where higher values indicate clearer community structure.

All the partition quality metrics defined above penalize misclassified inter-cluster positive and intra-cluster negative edges equally. However, empirical signed networks typically contain predominantly positive edges, with $m^+ \gg m^-$. For example, in the signed networks analyzed in this paper, positive edges account for 76% of all edges on average (see Table 3). Despite

their smaller number, negative edges provide crucial structural information in signed networks and should not be neglected. Consequently, an algorithm that simply ignores negative edges or treats them as positive edges, effectively disregarding edge signs, can appear to perform misleadingly well.

One way to mitigate this issue is to balance the penalization of misclassified positive and negative edges. In particular, the penalty of each misclassified positive and negative edge is weighted by $\sim 1/m^+$ and $\sim 1/m^-$, respectively. The normalization by m^+ and m^- ensures that positive and negative edges contribute equally overall, regardless of their relative abundance. We introduce normalized variants of the quality metrics, denoted by the superscript “*”, to distinguish them from their original counterparts. Note that when $m^+ = m^-$, the normalized and original definitions are equivalent.

The normalized variants of the metrics are defined as follows. The normalized frustration index $FI^* \in [0, 1]$ is

$$FI^* = \sum_{i < j} \frac{1}{2m^+} A_{ij}^+ [1 - \delta(c_i, c_j)] + \frac{1}{2m^-} A_{ij}^- \delta(c_i, c_j), \quad (4)$$

the normalized external–internal index $EI^* \in [-1, 1]$ is

$$EI^* = \frac{1}{2m^+} EI^+ - \frac{1}{2m^-} EI^-, \quad (5)$$

and the normalized modularity $Q^* \in [-1, 1]$ is

$$Q^* = \frac{1}{2m^+} Q^+ - \frac{1}{2m^-} Q^-. \quad (6)$$

In order to compare a detected partition with a reference partition, such as the planted community structure in synthetic benchmark graphs, we employ the mutual information between partitions adjusted for chance (Vinh et al., 2010). Let p_i denote the fraction of nodes in detected cluster i , r_j the fraction of nodes in reference community j , and p_{ij} the fraction of nodes belonging to both cluster i and community j . Then, the entropies of the two partitions are defined as $H_p = -\sum_i p_i \log p_i$ and $H_r = -\sum_j r_j \log r_j$, and their mutual information as $MI_{pr} = \sum_{ij} p_{ij} \log \frac{p_{ij}}{p_i r_j}$. Finally, the adjusted mutual information $AMI \leq 1$ is defined as

$$AMI = \frac{MI_{pr} - \mathbb{E}[MI_{pr}]}{\max\{H_p, H_r\} - \mathbb{E}[MI_{pr}]}, \quad (7)$$

where $\mathbb{E}[MI_{pr}]$ denotes the expected mutual information under the “permutation model”, in which partitions are generated randomly while preserving the number of clusters and the number of nodes in each cluster. Note that $AMI = 1$ when the partitions are identical up to a permutation of cluster labels, whereas $AMI = 0$ when their mutual information equals its expectation.

2.3. Post-hoc analysis

In this paper, we compare 13 community detection algorithms on 29 empirical signed networks. This results in 377 values for each performance metric, making a direct comparison cumbersome. An *ad hoc* approach is to average the values for each algorithm across all networks and compare their average performance. However, such an approach disregards the

Table 1: Hypothetical example of the performance of three algorithms on three networks according to a given metric. Although the algorithms have the same average metric value, their average ranks differ (see the main text for details).

Network	Algorithm	Metric ($\uparrow$)	Rank ($\downarrow$)
G_1	A	0.0	3
	B	0.5	2
	C	1.0	1
G_2	A	1.0	1
	B	0.5	2.5
	C	0.5	2.5
G_3	A	0.5	1.5
	B	0.5	1.5
	C	0.0	3

paired nature of the evaluations and may obscure substantial differences in algorithm performance across networks (Webb, 2000). In particular, metric values from different networks may not be directly comparable in magnitude, as the scale of a metric can depend on the properties of the underlying network.

As an illustration, consider a hypothetical comparison of algorithms A, B and C on networks G_1 , G_2 and G_3 . Table 1 shows the performance of the algorithms according to a metric for which higher values indicate better performance. The average performance of all three algorithms is 0.5. However, algorithm A has the best performance on networks G_2 and G_3 , whereas algorithm C has the worst performance. Thus, the relative performance of the algorithms varies across networks. Although their average metric values are identical, their average ranks differ.

A statistically principled approach is to rank the algorithms on each network, properly accounting for ties (see Table 1), and compare their average ranks rather than their average metric values (Demšar, 2006). Let N denote the number of networks, K the number of compared algorithms, and $r_{ij} \in [1, K]$ the rank of algorithm i on network j . The average rank of algorithm i is given by $r_i = \frac{1}{N} \sum_j r_{ij}$. The Friedman test compares the average ranks of the algorithms under the null hypothesis that all the algorithms are equivalent, with the test statistic

$$\chi_F^2 = \frac{12N}{K(K+1)} \left[\sum_i r_i^2 - \frac{K(K+1)^2}{4} \right], \quad (8)$$

which follows a χ^2 -distribution with $K - 1$ degrees of freedom (Friedman, 1937, 1940). Iman and Davenport (1980) correction accounts for the undesirably conservative nature of χ_F^2 and defines the statistic

$$F_{ID} = \frac{(N-1) \cdot \chi_F^2}{N(K-1) - \chi_F^2}, \quad (9)$$

which follows an F -distribution with $K - 1$ and $(K - 1)(N - 1)$ degrees of freedom.

If the null hypothesis is rejected, we can proceed with the post-hoc test of Nemenyi (1963). The performance of two algorithms i and j is considered statistically significantly different if the absolute difference between their average ranks $|r_i - r_j|$ is

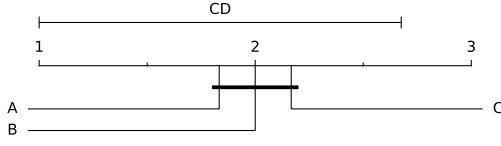


Figure 1: Nemenyi diagram showing the average ranks of algorithms for a hypothetical example (see the main text for details).

at least the critical difference

$$CD = q \sqrt{\frac{K(K+1)}{6N}}, \quad (10)$$

where q is the critical value of the Studentized range statistic divided by $\sqrt{2}$. In our hypothetical example, the average ranks are $r_A = 1.83$, $r_B = 2$, and $r_C = 2.17$, while the critical difference is $CD = 1.68$ at a p -value = 0.1. Therefore, pairwise differences in average ranks are not statistically significant, despite the algorithms having different average ranks.

The results of the Nemenyi test can be neatly visualized with a critical-difference diagram presented in Figure 1 (Demšar, 2006). The diagram plots the average ranks of the algorithms, with bold lines connecting algorithms that are *not* statistically significantly different. For realistic examples of Nemenyi diagrams, see Figures 5 and 6.

3. Fast algorithm

Label propagation is a fast algorithm for detecting community structure in unsigned networks (Raghavan et al., 2007). Traag and Šubelj (2023) recently proposed an even faster variant that provides additional benefits at no extra cost. In this paper, we extend the fast label propagation algorithm to signed networks. Earlier adaptations of the standard label propagation algorithm to signed networks include (Zarei et al., 2020; Hosseini-Pozveh et al., 2022). In the following, we first present the fast label propagation algorithm for unsigned networks and then its extensions to signed networks.

Label propagation for community detection is a heuristic approach based on the principle that each node should be assigned to the community to which most of its neighbors belong. The fast variant of the algorithm denoted FLPA operates as follows. Initially, each node $i \in V$ is assigned to a separate cluster by setting its cluster label to $c_i = i$. The nodes are randomly shuffled and added to the queue $Q \subseteq V$, which stores nodes whose cluster labels *may* change and therefore need to be processed. At each step of the algorithm, node $i \in Q$ is removed from the queue and a cluster label c is randomly selected from the set of most frequent labels in its neighborhood $\{c\} = \text{argmax}_c \sum_{j \in \Gamma_i} A_{ij} \delta(c_j, c)$. If the new label c is different from the current label c_i , the node updates $c_i = c$ and some of its neighbors Γ_i are pushed to the queue Q . Note that only neighbors $j \in \Gamma_i$ whose cluster labels differ from the new label $c_j \neq c$ may eventually update their labels and are therefore added to the queue Q , if they are not already in the queue. Once the

Require: undirected signed multigraph G

Ensure: node clustering $C = [c_i]_{i=1}^n$

```

1:  $C \leftarrow [1, 2, 3 \dots n]$ 
2:  $Q \leftarrow \text{shuffle } V$ 
3: while  $Q$  not empty do
4:    $i \leftarrow \text{pop } Q$ 
5:    $\{c\} \leftarrow \text{argmax}_c \sum_{j \in \Gamma_i} A_{ij} \delta(c_j, c)$ 
6:    $c \leftarrow \text{random}_c \{c\}$ 
7:   if  $c_i \neq c$  and  $\sum_{j \in \Gamma_i} A_{ij} \delta(c_j, c) \geq 0$  then
8:     for  $j \in \Gamma_i : j \notin Q$  do
9:       if  $c_j \neq c \wedge A_{ij} \geq 0$  or  $c_j \neq c_i \wedge A_{ij} < 0$  then
10:         $j \rightarrow \text{push } Q$ 
11:      end if
12:    end for
13:     $c_i \leftarrow c$ 
14:  end if
15: end while
16: return  $C$ 
```

Algorithm 1: Fast label propagation algorithm for signed networks (sFLPA).

queue Q is empty, each node $i \in V$ is guaranteed to have a cluster label that is among the most frequent labels in its neighborhood $c_i \in \{c\}$, thus the resulting labels $C = [c_i]_{i=1}^n$ are returned as the detected community structure. Since the algorithm is probabilistic, the random ordering of nodes in the queue Q and the random selection among the most frequent cluster labels $\{c\}$ may lead to different community structures in different runs.

We proceed by extending the fast label propagation algorithm to signed networks. The signed version is based on the principle that each node should be assigned to the community to which its neighbors connected by mostly positive edges belong and should *not* be assigned to the community to which its neighbors connected by mostly negative edges belong. This requires modifying the cluster label update rule and queue management to account for negative edges as well. Consider a node $i \in Q$ that is currently being processed. As before, a cluster label c is randomly selected from the set of candidate labels in its neighborhood $\{c\} = \text{argmax}_c \sum_{j \in \Gamma_i} A_{ij} \delta(c_j, c)$, where negative edges are now also accounted for through $A_{ij} = A_{ij}^+ - A_{ij}^-$. If the new label c differs from the current label c_i , the node will update its label $c_i = c$, but only if the edges connecting it to neighbors with label c have a nonnegative net sign $\sum_{j \in \Gamma_i} A_{ij} \delta(c_j, c) \geq 0$. In the latter case, two types of neighbors $j \in \Gamma_i$ may consequently change their labels and therefore need to be added to the queue Q , if they are not already in the queue. On the one hand, neighbors connected by mostly positive edges $A_{ij} \geq 0$ may adopt the new label c , provided that this differs from their current label $c_j \neq c$. On the other hand, neighbors connected by mostly negative edges $A_{ij} < 0$ may adopt the old label c_i , again provided that this differs from their current label $c_j \neq c_i$. All other aspects of the algorithm remain unchanged. The pseudocode of the fast label propagation algorithm for signed networks denoted sFLPA is shown in Algorithm 1.

Recall that empirical signed networks typically contain predominantly positive edges, with 76% of all edges being positive

Require: undirected signed multigraph G

Ensure: node clustering $C = [c_i]_{i=1}^n$

```

1:  $C \leftarrow [1, 2, 3 \dots n]$ 
2:  $Q \leftarrow \text{shuffle } V$ 
3: while  $Q$  not empty do
4:    $i \leftarrow \text{pop } Q$ 
5:    $\{c\} \leftarrow \text{argmax}_c \sum_{j \in \Gamma_i} \left( \frac{A_{ij}^+}{m^+} - \frac{A_{ij}^-}{m^-} \right) \delta(c_j, c)$ 
6:    $c \leftarrow \text{random}_c\{c\}$ 
7:   if  $c_i \neq c$  and  $\sum_{j \in \Gamma_i} \left( \frac{A_{ij}^+}{m^+} - \frac{A_{ij}^-}{m^-} \right) \delta(c_j, c) \geq 0$  then
8:     for  $j \in \Gamma_i : j \notin Q$  do
9:       if  $c_j \neq c \wedge \frac{A_{ij}^+}{m^+} \geq \frac{A_{ij}^-}{m^-}$  or  $c_j \neq c_i \wedge \frac{A_{ij}^+}{m^+} < \frac{A_{ij}^-}{m^-}$  then
10:         $j \rightarrow \text{push } Q$ 
11:      end if
12:    end for
13:     $c_i \leftarrow c$ 
14:  end if
15: end while
16: return  $C$ 

```

Algorithm 2: Fast label propagation algorithm for signed networks with normalized sign weights (sFLPA*).

in the networks in Table 3. This lead us to introduce normalized partition quality metrics in Section 2.2, which ensure that the overall contribution of positive edges equals that of negative edges. In particular, misclassified positive and negative edges are penalized by factors of $1/m^+$ and $1/m^-$, respectively.

Similarly, we can modify Algorithm 1 to effectively optimize the normalized metrics. Only two changes are needed. First, the set of candidate labels for node $i \in Q$ is changed to $\{c\} = \text{argmax}_c \sum_{j \in \Gamma_i} \left(\frac{A_{ij}^+}{m^+} - \frac{A_{ij}^-}{m^-} \right) \delta(c_j, c)$ (lines 5 and 7). Second, the conditions for adding neighbors $j \in \Gamma_i$ to the queue Q become $\frac{A_{ij}^+}{m^+} \geq \frac{A_{ij}^-}{m^-}$ and $\frac{A_{ij}^+}{m^+} < \frac{A_{ij}^-}{m^-}$ for $c_j \neq c$ and $c_j \neq c_i$, respectively (line 9). The pseudocode of the fast label propagation algorithm for signed networks incorporating normalized sign weights denoted sFLPA* is shown in Algorithm 2.

Algorithms 1 and 2 can also be used on weighted signed networks. In this case, A_{ij}^+ and A_{ij}^- are the sums of weights of positive and negative edges connecting nodes $i, j \in V$, while m^+ and m^- are the total weights of positive and negative edges, respectively.

4. Evaluation

In this section, we compare the fast label propagation algorithms FLPA, sFLPA, and sFLPA* presented in Section 3 with ten other community detection algorithms listed in Table 2. For a comprehensive evaluation, these include several algorithms for signed networks as well as popular algorithms for unsigned networks. Descriptions of all algorithms, along with their implementation specifics, are provided in Appendix A. We first evaluate the algorithms on synthetic signed graphs and then on empirical signed networks.

Table 2: Community detection algorithms compared in this paper. The prefix “s” denotes algorithms for signed networks that incorporate edge signs, whereas the remaining algorithms are for unsigned networks that ignore edge signs.

Algorithm	Description
FLPA	fast label propagation algorithm
DCSBM	degree-corrected stochastic block model
Walktrap	hierarchical clustering based on random walks
Louvain	Louvain algorithm optimizing modularity
Leiden	Leiden algorithm optimizing modularity
sHC	hierarchical clustering with cosine similarity
sRA	relocation algorithm using only single moves
sSG	spinglass method based on simulated annealing
sFLPA	FLPA for signed networks (Algorithm 1)
sFLPA*	sFLPA with normalized weights (Algorithm 2)
sDCSBM	DCSBM with Bernoulli edge covariates
sDCSBM ²	DCSBM for two-layered networks
sLouvain	Louvain algorithm for signed networks

4.1. Synthetic graphs

We start by evaluating the algorithms on synthetic signed graphs with planted community structure. We use Lancichinetti et al. (2008) graphs with $n = 1000$ nodes and the average node degree $\langle k \rangle = 10$. The node degrees and community sizes follow power-law distributions $p_k \sim k^{-3}$ and $p_s \sim s^{-2}$, respectively. The fraction of inter-community edges is controlled by the mixing parameter $\mu \in [0, 1]$. In the top row of Figure 2, the intra-community edges are assigned a positive sign and the inter-community edges are assigned a negative sign, thereby reinforcing the planted community structure. In the bottom row of Figure 2, all edge signs are flipped independently with probability 10%, introducing frustration into the planted structure. Note that, for clarity, we compare only the four best-performing algorithms on these graphs.

The left column of Figure 2 shows how well the algorithms recover the planted structure, as measured by the adjusted mutual information AMI . All algorithms reliably recover the structure up to $\mu \approx 0.4$ with $AMI > 0.9$ in unfrustrated graphs (top left plot) and up to $\mu \approx 0.3$ in frustrated graphs (bottom left plot). The stochastic block model sDCSBM performs best, whereas the fast label propagation algorithm sFLPA performs worst. The spinglass method sSG and modularity optimization sLouvain perform somewhere in between. However, the maximum difference in AMI among these algorithms remains below 0.08 for $\mu < 0.4$ and is therefore relatively small.

The right two columns of Figure 2 show the frustration index FI and the external-internal index EI of the detected community structure. The results follow similar trends as those observed for AMI , thereby justifying the use of these quality metrics to evaluate algorithms on empirical signed networks in Section 4.2, where no known community structure is available for comparison.

We proceed with evaluating the robustness of the algorithms for signed networks by applying them to synthetic signed graphs with no community structure. The algorithms should ideally not detect any structure in these graphs, for instance, by assigning every node to its own cluster or by assigning

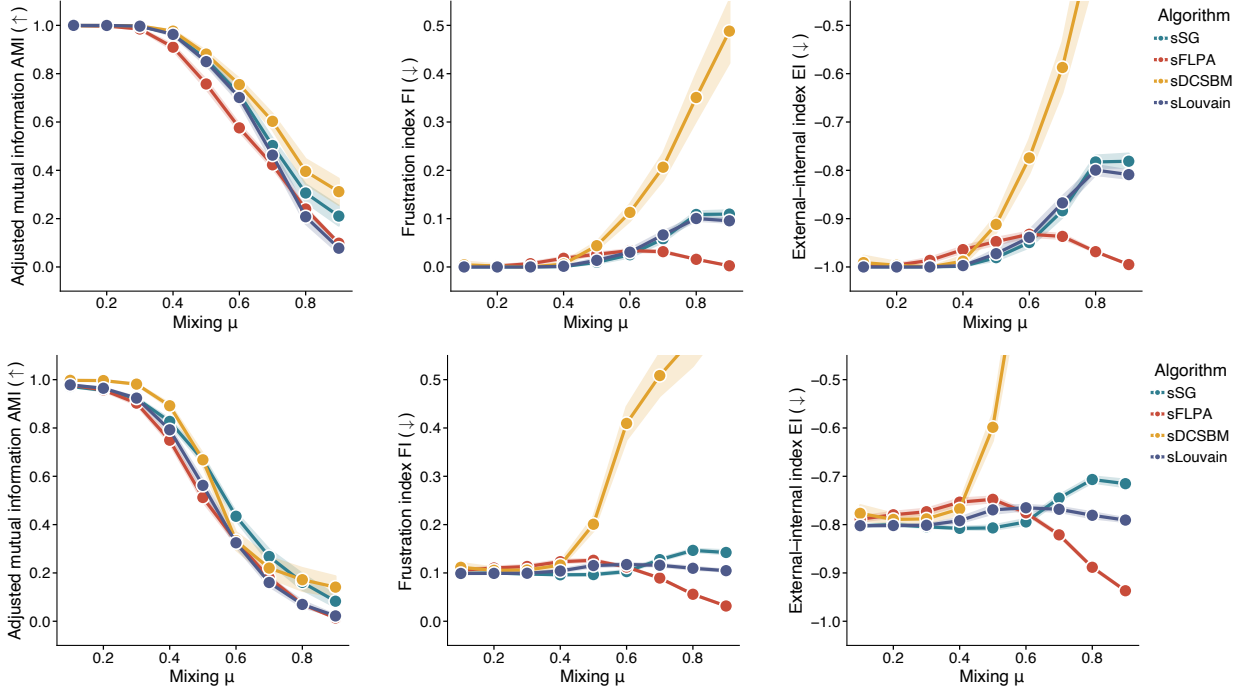


Figure 2: Performance of the best community detection algorithms on synthetic signed graphs with planted community structure. The fraction of inter-community edges is controlled by the mixing parameter μ . (top) Graphs with positive signs on intra-community edges and negative signs on inter-community edges. (bottom) Graphs in which edge signs are flipped independently with probability 10%. Results are averaged over 100 independent graph realizations.

all nodes to a single giant cluster. We consider three types of graphs, namely complete graphs K_n with $n = 100$ nodes, Erdős and Rényi (1959) random graphs $G(n, m)$ with $n = 1000$ nodes and $m = 5000$ edges, and flipped stochastic block model (SBM) graphs with $n = 1000$ nodes, an average node degree of $\langle k \rangle = 10$, and 10 equally sized planted blocks (Holland et al., 1983). Edge signs are generated according to a parameter $\gamma \in [0, 1]$ controlling the fraction of positive edges. For the complete and Erdős–Rényi graphs, $m^+ = \gamma m$ positive edges are selected uniformly at random, while the remaining edges are negative. For the flipped SBM graphs, $m^+ = \gamma m$ positive inter-block edges and $m^- = (1 - \gamma)m$ negative intra-block edges are used.

The left plot of Figure 3 shows the average size of the detected clusters in graphs with the edge-sign parameter $\gamma = 0.1$, where negative edges predominate. The fast label propagation algorithm sFLPA appropriately detects tiny clusters with an average of 1.8–3.2 nodes, as it groups together nodes connected by the few positive edges. A similar performance is observed for the relocation algorithm sRA and, in part, for the hierarchical clustering sHC. On the other hand, the spinglass method sSG and modularity optimization algorithm sLouvain perform poorly, detecting clusters with well more than 100 nodes in random graphs $G(n, m)$, which, for the latter, can be attributed to a known weakness of modularity optimization (Fortunato and Barthelemy, 2007; Peixoto, 2023). The stochastic block models sDCSBM and sDCSBM², however, exhibit a different but desirable behavior. They detect a single giant cluster in com-

plete graphs K_n and random graphs $G(n, m)$ ¹, and the planted blocks in flipped SBM graphs, which now effectively constitute an anti-community structure.

The right plot of Figure 3 shows the average fraction of nodes in the largest detected cluster in graphs with the edge-sign parameter $\gamma = 0.9$, where positive edges predominate. The fast label propagation algorithm sFLPA and the stochastic block models sDCSBM and sDCSBM² correctly identify a single giant cluster in all three graphs. The same performance is observed for the relocation algorithm sRA on complete graphs K_n . In contrast, all the other algorithms struggle to identify the expected single giant cluster, instead detecting multiple smaller clusters, despite the absence of any signed community structure in these graphs. Based on these experiments, we conclude that only sFLPA, sDCSBM, and sDCSBM² are robust enough to be reliably used on empirical signed networks that may lack any detectable community structure.

4.2. Empirical networks

We continue with a comprehensive evaluation of all algorithms in Table 2 on 29 empirical signed networks listed in Table 3. The networks span social, information, biological, and other domains and vary substantially in size and structure. They were collected from different sources and online repositories, with detailed descriptions provided in Appendix B.

The networks are divided into small networks with $n < 1500$ nodes and large networks with $n \geq 1500$ nodes. Due to their

¹Recall that complete graphs K_n contain only $n = 100$ nodes, whereas the other graphs contain $n = 1000$ nodes.

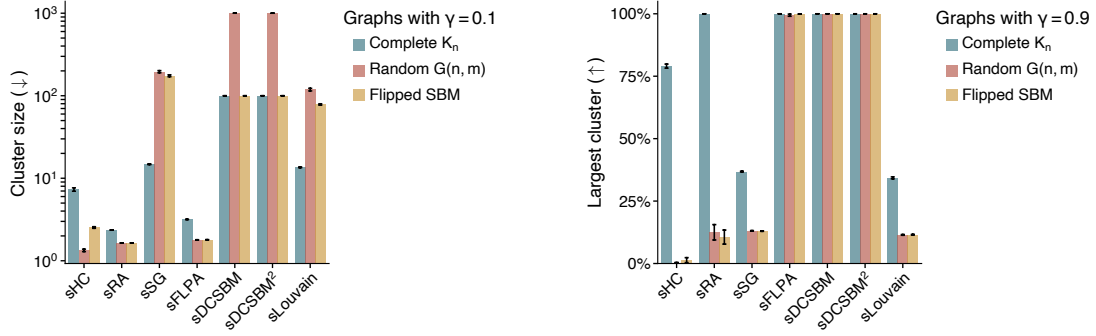


Figure 3: Performance of community detection algorithms on synthetic signed graphs with no community structure, where the edge signs are generated according to the parameter γ (see the main text for details). Results are averaged over 100 independent graph realizations.

Table 3: Statistics of empirical signed networks, including the number of nodes n and edges m , and the percentage of positive edges m^+/m . The networks are categorized as small with $n < 1500$ or large with $n \geq 1500$.

Network	Nodes n	Edges m	Positive
new-guinea-tribes	16	58	50.0%
newcomb3-15	17	119	57.1%
sampson-t4	18	103	54.4%
cow46	64	362	88.4%
spanish-school-t2	238	3 755	90.5%
EGFR-symm	329	779	66.1%
spanish-school-t1	409	8 557	85.3%
us-congress-h114	446	75 180	35.4%
spanish-school-t6	534	12 812	86.0%
org-conflict-17	551	2 931	87.9%
macrophage-symm	678	1 425	66.5%
yeast-symm	690	1 080	79.6%
ecoli-symm	1 461	3 215	58.4%
birdwatch-sg	2 987	441 986	63.1%
soc-bitcoin-alpha	3 783	24 186	93.6%
meneame-russia	4 548	105 414	89.9%
soc-bitcoin-otc	5 881	35 592	90.0%
chess	6 832	36 387	58.2%
elec	7 115	103 617	78.4%
meneame-politics	7 332	405 130	89.7%
compling-authors	7 493	169 289	91.5%
compling-papers	9 035	25 327	91.4%
wiki-rfa	11 259	185 540	77.8%
soc-reddit-body	35 776	286 561	92.6%
soc-reddit-title	54 075	571 927	89.3%
soc-Slashdot090221	82 140	549 202	77.4%
wikiconflict	116 717	2 915 501	43.4%
soc-epinions	131 580	840 799	85.3%
wiki-users	138 587	740 106	87.9%

higher computational cost, we apply the slower algorithms only to small networks, whereas the faster algorithms are evaluated on all networks. The left two plots in Figure 4 show the distributions of the number of nodes n and edges m , respectively. The rightmost plot in Figure 4 shows the distribution of the percentage of positive edges m^+/m . Notably, most networks contain predominantly positive edges, with positive edges accounting for 76% of all edges on average. This predominance of positive edges motivates including also algorithms designed for unsigned networks in the evaluation.

Results for all partition quality metrics introduced in Section 2 are reported across six tables in Appendix C. Given the large number of results, in the main text, we provide only Nemenyi diagrams showing the average ranks of the algorithms.

We first compare all algorithms on small networks. The top plot in Figure 5 shows the Nemenyi diagram for the frustration index FI , while the corresponding diagram for the external-internal index EI is identical, indicating that the two metrics yield the same ranking of the algorithms, although the relative differences between the algorithms still differ. The fast label propagation algorithm sFLPA achieves the lowest average rank of 2.62, followed by the modularity optimization sLouvain with 2.81, while the hierarchical clustering sHC and the relocation algorithm sRA both have an average rank of 4.15. The best-performing algorithms for unsigned networks are the modularity optimization Louvain and the fast label propagation algorithm FLPA, with average ranks of 7.08 and 7.19, respectively. However, the difference between their average ranks is not statistically significant at a p -value of 0.1. In contrast, sFLPA and sLouvain are statistically significantly better than the remaining algorithms for signed networks, including the spinglass method sSG and the stochastic block models sDCSBM and sDCSBM². It should be noted that the latter are designed to detect the block structure most strongly expressed in the network rather than specifically community structure (Peixoto, 2019). Thus, sDCSBM and sDCSBM² may have identified a type of structure other than community structure, which is beyond the scope of this paper.

The middle plot in Figure 5 shows the Nemenyi diagram for the normalized frustration index FI^* , while the diagram for the normalized external-internal index EI^* is again identical. As expected, the fast label propagation algorithm incorporating

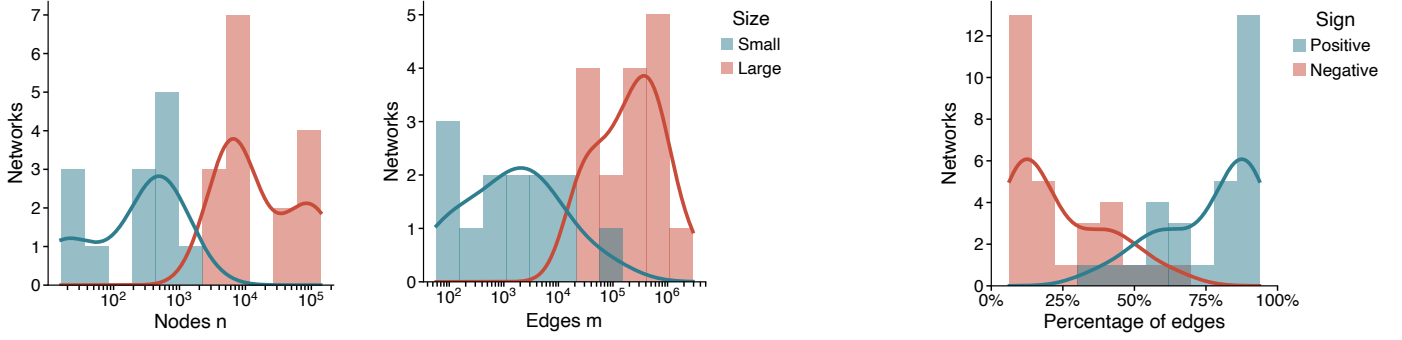


Figure 4: (left) Distributions of the number of nodes n and edges m in empirical signed networks, shown separately for small networks with $n < 1500$ and large networks with $n \geq 1500$. (right) Distributions of the percentages of positive edges m^+/m and negative edges m^-/m .

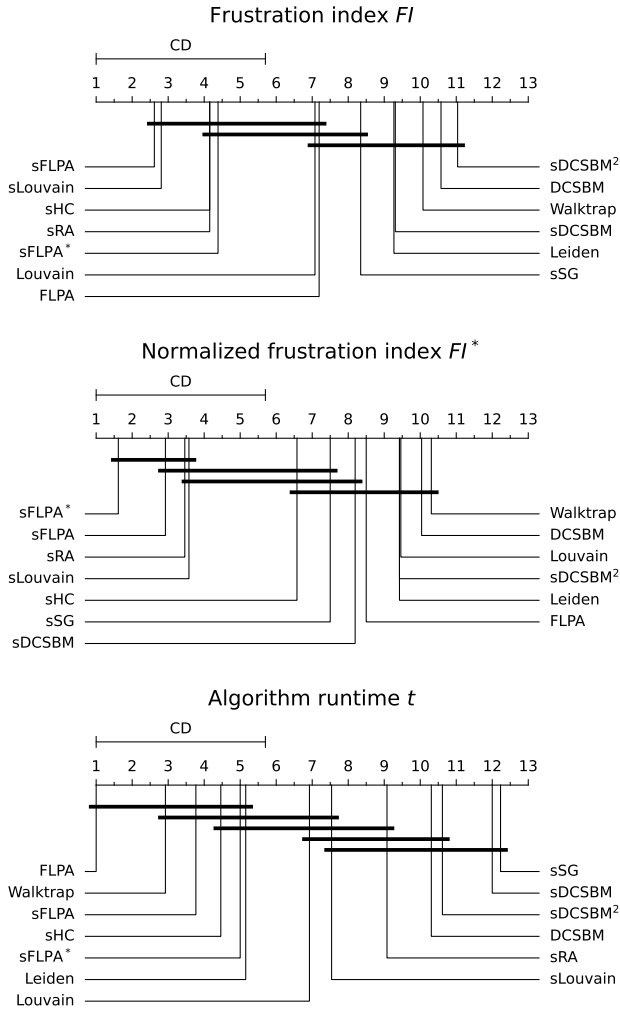


Figure 5: Nemenyi diagrams showing the average ranks of community detection algorithms on small empirical signed networks with $n < 1500$ nodes. Average ranks are computed over 100 runs, with bold lines connecting algorithms whose differences are not statistically significant at a p -value = 0.1.

normalized sign weights sFLPA* performs best, with an average rank of 1.62. At a p -value = 0.1, the algorithm is not statistically significantly better than the unnormalized variant sFLPA with an average rank of 2.92, the relocation algorithm sRA with 3.46, or the modularity optimization sLouvain with 3.58. Nevertheless, sFLPA* is statistically significantly better than all remaining algorithms for signed networks, including sHC, sSG, sDCSBM, and sDCSBM².

The bottom plot in Figure 5 shows the Nemenyi diagram for the running time of the algorithms. The fast label propagation algorithm for unsigned networks FLPA, which was designed for computational efficiency (Traag and Šubelj, 2023), ranks first on all networks. The signed variant sFLPA is the fastest among the algorithms for signed networks, with an average rank of 3.77, followed by the hierarchical clustering sHC with 4.46, the normalized variant sFLPA* with 5.00, and the modularity optimization sLouvain with 7.54. The differences between these algorithms are not statistically significant at a p -value of 0.1. Nevertheless, sFLPA is statistically significantly faster than the remaining algorithms for signed networks, including the relocation algorithm sRA, the stochastic block models sDCSBM and sDCSBM², and the spinglass method sSG.

We conclude that, at least on small networks, the fast label propagation algorithms for signed networks sFLPA and sFLPA* outperform the other algorithms in terms of the quality of the detected signed community structure, as measured by the frustration index FI and the external-internal index EI , both in their standard and normalized forms, while also achieving the lowest running times. For example, on the largest of these networks, *ecoli-symm* with $n = 1461$ nodes, sFLPA and sFLPA* take 0.015 and 0.017 seconds on average, respectively, whereas sSG takes 7.741 seconds on average.

We continue by comparing only the faster algorithms on all networks. These include the fast label propagation algorithms sFLPA and sFLPA*, and the modularity optimization sLouvain. We also include their unsigned counterparts FLPA, Louvain and Leiden. Since the Nemenyi diagrams for the frustration indices FI and FI^* are again identical to the corresponding diagrams for the external-internal indices EI and EI^* , we show and discuss only the former in Figure 6.

The top plot in Figure 6 shows the Nemenyi diagram for the frustration index FI . The fast label propagation algorithm

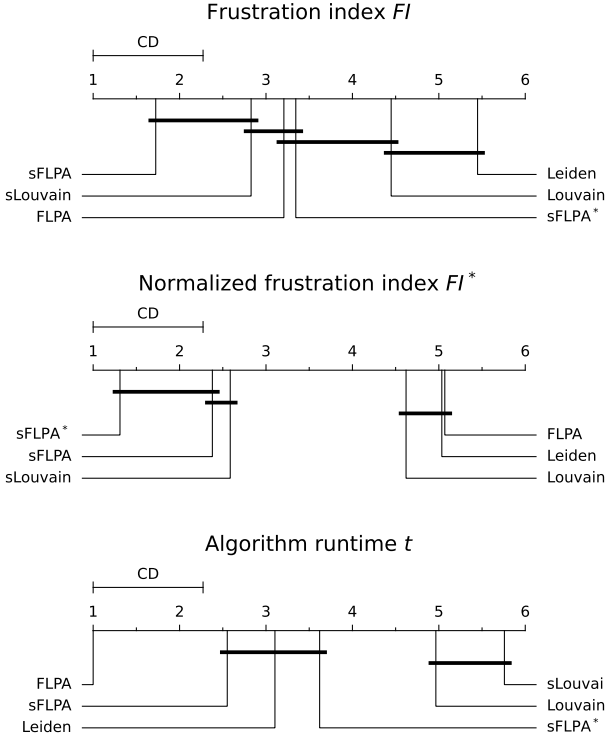


Figure 6: Nemenyi diagrams showing the average ranks of the faster community detection algorithms on all empirical signed networks. Average ranks are computed over 100 runs, with bold lines connecting algorithms whose differences are not statistically significant at a p -value = 0.1.

sFLPA achieves the best performance, with an average rank of 1.72, followed by the modularity optimization sLouvain with 2.83. The differences between these two algorithms are not statistically significant at a p -value = 0.1. On the other hand, sFLPA is statistically significantly better than its unsigned variant FLPA, which has an average rank of 3.21, and the normalized variant sFLPA* with 3.34. The modularity optimization algorithms for unsigned networks, namely Louvain and Leiden, have the worst performance with 4.45 and 5.45, respectively.

The middle plot in Figure 6 shows the Nemenyi diagram for the normalized frustration index FI^* . As expected, the fast label propagation algorithm incorporating normalized sign weights sFLPA* has the lowest average rank of 1.31. At a p -value = 0.1, the algorithm is statistically significantly better than the modularity optimization sLouvain with 2.59. Furthermore, all algorithms for signed networks are statistically significantly better than their unsigned counterparts.

Finally, the bottom plot in Figure 6 shows the Nemenyi diagram for the running time of the algorithms. As before, the fast label propagation algorithm for unsigned networks FLPA ranks first across all networks. The signed variants sFLPA and sFLPA* rank second and fourth, with average ranks of 2.55 and 3.62, respectively. Both sFLPA and sFLPA* are statistically significantly faster than the modularity optimization sLouvain at a p -value = 0.1. For example, on the wikiconflict network with more than a hundred thousand nodes, sFLPA and sFLPA* take 19.2 and 15.8 seconds on average, respectively, whereas sLouvain takes 227.7 seconds on average, making it approximately 12–14 $\times$ slower than the label propagation algorithms.

Therefore, our conclusions regarding overall algorithm performance on empirical networks remain unchanged also for the larger networks. The fast label propagation algorithms outperform the other algorithms in terms of the quality of the detected structure and running time. The algorithms provide substantial benefits over existing alternatives at no additional cost and should thus be the preferred choice in practice.

4.3. Der Standard network

Despite the fast label propagation algorithm proving to be the fastest algorithm compared in this paper, the algorithm was developed for sparse networks, and it might not converge in a reasonable time on a very dense signed network with similar numbers of positive and negative edges. Previous literature has suggested either limiting the number of steps of the algorithm (Xie et al., 2011; Ferrara, 2012) or stopping it when the majority of nodes have already been assigned their final cluster (Raghavan et al., 2007; Leung et al., 2009). We find such *ad hoc* approaches undesirable since the algorithm then no longer guarantees that each node is assigned to a cluster with the highest net sign in its neighborhood.

A better and recommended solution is to adjust the initialization of the clusters in step 1 of Algorithms 1 and 2. Instead of initially assigning each node $i \in \{1, \dots, n\}$ to a separate cluster by setting its cluster label to $c_i = i$, c_i is randomly sampled from some finite set of cluster labels. In fact, in most empirical networks, the number of detected clusters is usually much smaller than the number of nodes n . Therefore, choosing the

Table 4: Statistics of clusterings detected in Der Standard network using different algorithms, including the number of clusters, the fraction of nodes in the largest cluster, the frustration index FI , the external–internal index EI , and the algorithm runtime t . Statistics are averaged over 100 runs, with the best values highlighted in bold.

Algorithm	Clusters	Largest	FI ($\downarrow$)	EI ($\downarrow$)	t [sec]
sLouvain	3.0	50.6%	0.352	−0.296	213.4
	2.0	71.9%	0.319	−0.362	28.0
sFLPA	3.0	63.8%	0.285	−0.430	32.6
	5.0	57.2%	0.257	−0.486	46.7
	10.0	52.0%	0.232	−0.535	77.0

number of initial cluster labels from the range $[2, \sqrt{n}]$ is a reasonable choice, which is also supported by theoretical results on the detectability of communities (Peixoto, 2013) and the resolution limit of community detection algorithms (Fortunato and Barthelemy, 2007). Note that the number of initial cluster labels does not guarantee that the same number of clusters will be detected, since some cluster labels may disappear during propagation.

As an illustration, we consider a network compiled from a major Austrian online newspaper called Der Standard² (Fraxanet et al., 2024). Nodes represent regular users of the newspaper’s discussion forums, while positive and negative edges represent inferred positive and negative relationships between users based on their interactions in news discussions between 2014 and 2022. The network contains $n = 14\,827$ nodes and $m = 5\,559\,385$ signed edges, of which only 59% have a positive sign. The average node degree is $\langle k \rangle = 749.9$, which is by far the highest among the networks analyzed in this paper.

The original authors studied the division of the network into two balanced clusters, whereas the modularity optimization algorithm sLouvain detects 3.0 clusters on average. Therefore, for the fast label propagation algorithm sFLPA, we set the initial number of clusters to 2, 3, 5 or 10.

The results are shown in Table 4. All algorithms detect one large cluster containing more than half of the nodes. However, sFLPA detects markedly better signed community structure than sLouvain, as indicated by its lower frustration index FI and external–internal index EI . Furthermore, sFLPA is substantially faster than sLouvain. In particular, when both algorithms detect the same number of clusters, sFLPA is 6.5× faster than sLouvain. Notice also that increasing the number of clusters for sFLPA decreases both indices FI and EI , while increasing the running time. Therefore, in practice, choosing a larger number of clusters can improve the quality of the detected structure at the cost of increased running time.

5. Conclusions

Community detection is a frequent task in network analysis. A wide variety of approaches exist, offering different interpretations of what constitutes a community and corresponding algorithms for identifying them (Fortunato and Hric, 2016). In

signed networks with positive and negative edges, the goal is to identify a balanced partition, in which positive edges lie within clusters and negative edges lie between clusters. However, considerably fewer algorithms have been developed for community detection in signed networks (Diaz-Diaz et al., 2025).

In this paper, we extend the fast label propagation algorithm of Traag and Šubelj (2023) to signed networks. Extensive experiments on synthetic graphs and 30 empirical signed networks show that the proposed algorithm detects high-quality signed community structure while requiring substantially less computational time than existing approaches. In particular, it outperforms existing algorithms in terms of the quality of the detected structure, as measured by the frustration index and external–internal index, while achieving running times comparable to those of fast algorithms designed for unsigned networks. These results demonstrate that fast label propagation provides an effective and computationally efficient approach to community detection in signed networks, offering substantial practical advantages over existing alternatives.

In addition to the proposed algorithm, this paper introduces normalized variants of established partition quality metrics for signed networks. The normalized variants of the frustration index and external–internal index balance the contributions of positive and negative edges, enabling more meaningful evaluation of community structure in networks with highly unequal proportions of edge signs. Together with the proposed post-hoc statistical analysis based on average algorithm ranks visualized using critical-difference diagrams (Demšar, 2006), we provide a systematic framework for comparing community detection algorithms across diverse signed networks.

Finally, to facilitate reproducibility and future comparisons, we make the complete source code and experimental results publicly available in a GitHub repository. The repository includes implementations of the proposed algorithms and partition quality metrics, scripts for the post-hoc statistical analysis, and all results reported in this paper. The repository thus provides a comprehensive resource for reproducing and extending systematic comparisons of community detection algorithms on signed networks.

Several directions remain for future work. Although the proposed algorithms can be naturally extended to weighted signed networks, their performance in this setting remains to be evaluated. Furthermore, extending the proposed algorithms to directed signed networks requires a conceptual choice regarding what constitutes a community, as several different notions of community structure have been proposed for directed networks (Malliaros and Vazirgiannis, 2013). Correlation networks represent another common class of signed networks not considered in our study (Gómez et al., 2009), and evaluating the proposed algorithms on such networks is left for future work.

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²<https://dataverse.bsc.es/dataset.xhtml?persistentId=perma:BSC/P32CXW>

terpretation of the results, writing of the manuscript, or decision to submit the paper for publication.

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Appendix A. Algorithms

All community detection algorithms listed in Table 2 were implemented and executed in Python 3.14, using implementations developed by us, by the algorithms’ original authors, or provided by established third-party libraries. We primarily use NetworkX 3.6 for graph representation and generation (Hagberg et al., 2008), and CDlib 0.4 for partition quality metrics and comparison (Rossetti et al., 2019). To ensure reproducibility, the source code of the entire evaluation framework will be provided in a public GitHub repository. All experiments were performed on a MacBook Pro equipped with an Apple M4 Pro processor, 48 GB of memory, and running macOS 26.6. All algorithms were executed on the CPU without GPU acceleration. Note that some algorithm implementations require a non-standard graph representation. The corresponding graph transformations were performed beforehand and are not included in the algorithm running time. Unless explicitly stated otherwise, we use the default parameters of the existing implementations.

We provide our own implementations of the fast label propagation algorithms FLPA, sFLPA and sFLPA*. The implementations strictly follow the pseudocodes presented in Section 3. Additionally, we provide our own implementation of the relocation algorithm sRA (Doreian and Mrvar, 1996, 2009), which was the first widely used algorithm for community detection in signed networks. The algorithm iteratively moves nodes between communities to minimize the frustration index in Equation 1. For efficiency, we implement only single-node moves.

Louvain is a well-known modularity optimization algorithm for unsigned networks that consists of greedy local optimization and hierarchical aggregation (Blondel et al., 2008). Leiden is an improved version of Louvain that typically achieves faster convergence through an additional refinement phase and guarantees well-connected communities (Traag et al., 2019). For both algorithms, we use the CDlib 0.4 wrappers around implementations provided by external Python libraries. sLouvain is an adaptation of Louvain for signed networks that optimizes the signed modularity defined in Equation 3 by operating on a two-layer network comprising positive and negative edges (Pougué-Biyong and Lambiotte, 2024). We use the implementation provided by the original authors³.

DCSBM is a degree-corrected stochastic block model for unsigned networks that minimizes the description length using a Markov chain Monte Carlo algorithm (Peixoto, 2019). The model is constrained to assortative structures through the “planted partition model” (Zhang and Peixoto, 2020). After fitting the model, we perform an additional 100 merge-split sweeps to refine the detected structure (Peixoto, 2020). We use an efficient implementation in graph-tool 2.98 (Peixoto, 2014).

³<https://github.com/lejohnjohn/signed-louvain>

Note that the library does not provide a dedicated implementation for signed networks. Therefore, we consider the following two approaches suggested by the library’s author. The first approach sDCSBM models edge signs as Bernoulli edge covariates (Peixoto, 2018). The second approach sDCSBM² operates on a two-layer network comprising positive and negative edges (Peixoto, 2015).

sHC performs hierarchical agglomerative clustering with average linkage (Manning et al., 2008). Each node is represented by the corresponding row or column of the adjacency matrix and the distance between two nodes is defined using cosine similarity. The resulting clustering dendrogram is cut at 60% of the maximum pairwise distance between the nodes to obtain the final clusters. This threshold was selected since it provided good results in preliminary experiments. We use the hierarchical clustering implementation provided by SciPy 1.17 (Virtanen et al., 2020).

sSG detects communities using a spinglass model with an energy function that combines separate terms for positive and negative edges (Reichardt and Bornholdt, 2006; Traag and Bruggeman, 2009). The community assignment is obtained by minimizing the resulting Hamiltonian using simulated annealing. We use the implementation provided by python-igraph 1.0 (Csardi and Nepusz, 2006), which requires setting an upper limit on the number of communities. As suggested by the authors, we set this limit to $\sqrt{n}$, where n is the number of nodes in the network. Additionally, the implementation requires a connected network. Therefore, for a disconnected network, the algorithm is applied to each connected component separately.

Walktrap is a community detection algorithm for unsigned networks based on network dynamics (Pons and Latapy, 2006). The algorithm performs short random walks and uses their trajectories to merge nodes into communities in a bottom-up manner. We use the CDlib 0.4 wrapper around the implementation provided by python-igraph 1.0.

Appendix B. Networks

In this appendix, we provide descriptions of the empirical signed networks listed in Table 3. All networks are represented as unweighted, undirected, signed multigraphs after removing self-loops and isolated nodes. The datasets used to compile the networks and the corresponding parsers will be available in a public GitHub repository.

new-guinea-tribes⁴ is a social network of friendships among sixteen tribes of the Gahuku–Gama alliance structure in New Guinea (Read, 1954). Positive edges indicate friendship, while negative edges indicate enmity.

newcomb3-15⁵ is a social network of seventeen students who became roommates at the University of Michigan in 1956 (Newcomb, 1961). Each student ranked the others, with positive edges assigned to the first four ranks and negative edges to the last three ranks.

sampson-t4⁶ is a social network of eighteen novices preparing to join a monastic order at a New England monastery (Sampson, 1968). Each novice identified their most and least liked peers, represented by positive and negative edges, respectively.

cow46⁷ is an international relations network of countries in 1946 (Doreian and Mrvar, 2015). Positive edges indicate joint membership in alliances, unions, or agreements, while negative edges indicate wars, conflicts, border disputes, or sharp disagreements.

spanish-school-t1, spanish-school-t2, and spanish-school-t6⁸ are social networks encoding friendship relations among students in three Spanish high schools (Ruiz-García et al., 2023). Edges represent friendship relations, with positive and negative signs indicating good and bad relationships, respectively.

EGFR-symm, macrophage-symm, yeast-symm, and ecoli-symm⁹ are biological networks representing molecular interactions in the EGFR pathway, a macrophage, yeast, and *E. coli*, respectively (Iacono et al., 2010). Positive and negative edges represent activating and inhibiting interactions, respectively, while the original directed networks were symmetrized.

us-congress-h114¹⁰ is a social network of members of the U.S. House of Representatives during the 114th Congress (Neal, 2020). Positive edges indicate a significant tendency to co-sponsor bills, while negative edges indicate a significant tendency not to co-sponsor bills.

org-conflict-17 is an organizational network of employees of a company operating in Belarus, Poland, and Russia (Chernenko, 2019). Positive edges represent employees recommending colleagues as people they would like to work with, while negative edges represent reported conflicts between employees.

birdwatch-sg¹¹ is a social network of users participating in Twitter’s Birdwatch pilot (Pougué-Biyong et al., 2023). Positive and negative edges indicate helpful and not helpful ratings, respectively, of notes written by other participants.

soc-bitcoin-alpha¹² and soc-bitcoin-otc¹³ are who-trusts-whom networks of members on the Bitcoin Alpha and Bitcoin OTC platforms, respectively (Kumar et al., 2016). Members rate others on a scale from –10 indicating total distrust to 10 indicating total trust. Positive and negative edges represent positive and negative ratings, respectively.

meneame-russia and meneame-politics¹⁴ are social networks of users interacting on the Spanish social media platform

⁶<http://vlado.fmf.uni-lj.si/pub/networks/data/esna/sampson.htm>

⁷<http://mrvar.fdv.uni-lj.si/pajek/SVG/CoW/default.htm>

⁸<https://zenodo.org/records/7647000#.Y-5eDtLMJH4>

⁹<https://www.sissa.it/fa/altafini/papers/IaA109>

¹⁰https://networks.skewed.de/net/us_congress

¹¹<https://github.com/lejohnyjohn/learning-stance-embeddings-from-signed-social-graphs>

¹²<http://snap.stanford.edu/data/soc-sign-bitcoin-alpha.html>

¹³<http://snap.stanford.edu/data/soc-sign-bitcoin-otc.html>

¹⁴<https://zenodo.org/records/15682068>

⁴https://networks.skewed.de/net/new_guinea_tribes

⁵<http://mrvar.fdv.uni-lj.si/sola/info4/andrej/signed.htm>

Menéame, focusing on the Russia–Ukraine war and political topics, respectively (Candellone et al., 2025). Positive and negative edges represent upvotes and downvotes of other users’ comments, respectively.

`chess`¹⁵ is a social network of chess players representing games played among the world’s top players (Kaggle, 2017). Each edge represents a game between two players, with positive and negative signs indicating wins by the white and black player, respectively. Drawn games are excluded.

`elec`¹⁶ is a social network of Wikipedia editors representing votes in Request for Adminship elections from a 2008 snapshot (Leskovec et al., 2010a). An edge indicates that one editor voted on another, with positive and negative signs indicating votes for and against, respectively.

`compling-authors` and `compling-papers`¹⁷ are citation networks of authors and papers in computational linguistics, respectively (Kumar, 2016). Positive and negative edges represent citations classified as endorsement and criticism, respectively, based on the text surrounding each citation.

`wiki-rfa`¹⁸ is a social network of Wikipedia members representing votes in Requests for Adminship elections from 2003 to 2013 (West et al., 2014). An edge indicates that one member voted on another, with positive and negative signs indicating votes for and against, respectively.

`soc-reddit-body` and `soc-reddit-title`¹⁹ are information networks of subreddits on Reddit, representing hyperlinks extracted from the body and title of posts, respectively, from 2014 to 2017 (Kumar et al., 2018). Positive and negative edges indicate neutral or positive and explicitly negative sentiment towards the target subreddit, respectively.

`soc-Slashdot090221`²⁰ is a social network of users on the Slashdot technology-related news website collected in 2009 (Leskovec et al., 2010b). Positive and negative edges represent friend and foe declarations between users, respectively.

`wikiconflict`²¹ is a social network representing interactions among editors of the English Wikipedia in 2009 (Brandes et al., 2009). Edges represent edit conflicts between editors, with positive and negative signs indicating agreement and disagreement, respectively.

`soc-epinions`²² is a who-trusts-whom network of users on the Epinions consumer review website (Leskovec et al., 2010b). Positive and negative edges represent trust and distrust relationships, respectively.

`wiki-users`²³ is a social network derived from interactions between editors of the English Wikipedia, based on the edit histories of 563 pages related to politics (Maniu et al., 2011). Positive and negative edges indicate trust or similarity and distrust

or disagreement, respectively.

Appendix C. Results

Tables C.5 to C.10 show the partition quality metrics of the clusterings detected by the community detection algorithms in Table 2 on the empirical signed networks listed in Table 3. The descriptions of the algorithms and networks are provided in Appendix A and Appendix B, respectively. To ensure reproducibility, all experimental results will be available in a public GitHub repository.

¹⁵<https://networks.skewed.de/net/chess>

¹⁶<https://networks.skewed.de/net/elec>

¹⁷<https://cs.stanford.edu/~srijan/citations>

¹⁸https://networks.skewed.de/net/wiki_rfa

¹⁹<http://snap.stanford.edu/data/soc-RedditHyperlinks.html>

²⁰<https://snap.stanford.edu/data/soc-sign-Slashdot090221.html>

²¹<https://networks.skewed.de/net/wikiconflict>

²²<https://snap.stanford.edu/data/soc-sign-epinions.html>

²³https://networks.skewed.de/net/wiki_users

Table C.5: Statistics of clusterings detected in empirical signed networks using different algorithms, including the number of clusters, the fraction of nodes in the largest cluster, the frustration index FI , the normalized frustration index FI^* , the external–internal index EI , the normalized external–internal index EI^* , the signed modularity Q , the normalized signed modularity Q^* , and the algorithm runtime t . Statistics are averaged over 100 runs, with the best values highlighted in bold. (1/6)

Network	Algorithm	Clusters	Largest	FI ($\downarrow$)	FI^* ($\downarrow$)	EI ($\downarrow$)	EI^* ($\downarrow$)	Q ($\uparrow$)	Q^* ($\uparrow$)	t [sec]
new-guinea-tribes	sHC	3.0	43.8%	0.034	0.034	-0.931	-0.931	0.431	0.431	2.2×10^{-4}
	sRA	3.5	43.8%	0.051	0.051	-0.899	-0.899	0.410	0.410	4.1×10^{-4}
	sSG	3.0	43.8%	0.034	0.034	-0.931	-0.931	0.431	0.431	1.6×10^{-2}
	sFLPA	3.1	43.8%	0.037	0.037	-0.927	-0.927	0.428	0.428	2.6×10^{-4}
	sFLPA*	3.0	43.8%	0.036	0.036	-0.928	-0.928	0.429	0.429	2.9×10^{-4}
	sDCSBM	2.0	75.0%	0.121	0.121	-0.759	-0.759	0.308	0.308	8.9×10^{-2}
	sDCSBM ²	1.0	100.0%	0.500	0.500	0.000	0.000	0.000	0.000	5.1×10^{-2}
	sLouvain	3.0	43.8%	0.034	0.034	-0.931	-0.931	0.431	0.431	8.4×10^{-4}
	FLPA	1.0	99.6%	0.500	0.500	-0.000	-0.000	0.001	0.001	10.0×10^{-5}
	DCSBM	1.0	100.0%	0.500	0.500	0.000	0.000	0.000	0.000	3.6×10^{-2}
	Walktrap	3.0	50.0%	0.569	0.569	0.138	0.138	-0.000	-0.000	2.7×10^{-4}
	Louvain	2.9	46.4%	0.463	0.463	-0.075	-0.075	0.017	0.017	5.4×10^{-4}
	Leiden	3.0	45.2%	0.465	0.465	-0.069	-0.069	0.011	0.011	7.6×10^{-4}
newcomb3-15	sHC	3.0	76.5%	0.227	0.225	-0.546	-0.549	0.089	0.106	2.4×10^{-4}
	sRA	4.1	57.5%	0.226	0.206	-0.549	-0.588	0.166	0.173	7.5×10^{-4}
	sSG	3.0	41.2%	0.319	0.294	-0.361	-0.412	0.193	0.198	2.2×10^{-2}
	sFLPA	4.0	61.9%	0.213	0.197	-0.574	-0.606	0.155	0.163	4.5×10^{-4}
	sFLPA*	4.0	59.3%	0.218	0.200	-0.564	-0.601	0.164	0.172	5.1×10^{-4}
	sDCSBM	2.0	76.5%	0.227	0.228	-0.546	-0.544	0.113	0.132	1.1×10^{-1}
	sDCSBM ²	1.0	100.0%	0.429	0.500	-0.143	0.000	0.000	0.000	5.8×10^{-2}
	sLouvain	3.3	38.7%	0.343	0.319	-0.314	-0.363	0.185	0.190	1.3×10^{-3}
	FLPA	1.0	100.0%	0.429	0.500	-0.143	0.000	0.000	0.000	1.1×10^{-4}
	DCSBM	1.0	100.0%	0.429	0.500	-0.143	0.000	0.000	0.000	4.3×10^{-2}
	Walktrap	2.0	64.7%	0.403	0.426	-0.193	-0.147	0.088	0.074	3.7×10^{-4}
	Louvain	2.8	44.8%	0.488	0.488	-0.024	-0.024	0.062	0.050	8.0×10^{-4}
	Leiden	2.7	49.1%	0.414	0.417	-0.171	-0.166	0.094	0.079	7.6×10^{-4}
sampson-t4	sHC	2.0	61.1%	0.155	0.163	-0.689	-0.673	0.343	0.337	2.4×10^{-4}
	sRA	3.7	38.7%	0.130	0.121	-0.741	-0.759	0.382	0.374	5.6×10^{-4}
	sSG	3.0	38.9%	0.117	0.107	-0.767	-0.786	0.396	0.391	2.0×10^{-2}
	sFLPA	3.0	38.9%	0.108	0.103	-0.783	-0.794	0.395	0.387	4.0×10^{-4}
	sFLPA*	3.1	39.1%	0.112	0.106	-0.777	-0.788	0.393	0.385	4.6×10^{-4}
	sDCSBM	2.0	61.1%	0.155	0.163	-0.689	-0.673	0.343	0.337	1.1×10^{-1}
	sDCSBM ²	1.0	100.0%	0.456	0.500	-0.087	0.000	0.000	0.000	5.7×10^{-2}
	sLouvain	3.0	38.9%	0.117	0.107	-0.767	-0.786	0.396	0.391	1.1×10^{-3}
	FLPA	1.0	100.0%	0.456	0.500	-0.087	0.000	0.000	0.000	1.1×10^{-4}
	DCSBM	1.0	100.0%	0.456	0.500	-0.087	0.000	0.000	0.000	4.3×10^{-2}
	Walktrap	2.0	55.6%	0.301	0.313	-0.398	-0.375	0.206	0.195	3.5×10^{-4}
	Louvain	2.9	44.2%	0.397	0.398	-0.206	-0.205	0.117	0.105	7.3×10^{-4}
	Leiden	3.0	34.3%	0.385	0.385	-0.230	-0.230	0.132	0.118	7.1×10^{-4}
cow46	sHC	2.0	85.9%	0.086	0.245	-0.829	-0.510	0.006	0.027	4.3×10^{-4}
	sRA	16.9	33.2%	0.065	0.089	-0.870	-0.821	0.394	0.222	1.3×10^{-2}
	sSG	7.0	32.8%	0.083	0.109	-0.834	-0.782	0.432	0.294	1.8×10^{-1}
	sFLPA	16.8	33.0%	0.064	0.087	-0.872	-0.826	0.397	0.226	1.2×10^{-3}
	sFLPA*	17.7	32.8%	0.074	0.062	-0.853	-0.875	0.388	0.245	1.4×10^{-3}
	sDCSBM	5.0	35.9%	0.169	0.344	-0.663	-0.313	0.368	0.094	1.3×10^{-1}
	sDCSBM ²	5.1	34.0%	0.196	0.366	-0.608	-0.268	0.343	0.065	8.5×10^{-2}
	sLouvain	7.0	32.0%	0.086	0.123	-0.827	-0.753	0.430	0.286	3.6×10^{-3}
	FLPA	7.2	34.2%	0.114	0.306	-0.772	-0.388	0.360	0.069	4.2×10^{-4}
	DCSBM	8.6	41.4%	0.383	0.485	-0.234	-0.031	0.280	0.069	4.3×10^{-2}
	Walktrap	8.0	31.2%	0.133	0.313	-0.735	-0.374	0.380	0.083	9.5×10^{-4}
	Louvain	6.0	31.2%	0.135	0.328	-0.731	-0.344	0.382	0.085	2.6×10^{-3}
	Leiden	6.0	31.2%	0.130	0.312	-0.740	-0.375	0.387	0.102	1.8×10^{-3}

Table C.6: (continued) Statistics of clusterings detected in empirical signed networks using different algorithms. (2/6)

Network	Algorithm	Clusters	Largest	FI ($\downarrow$)	FI^* ($\downarrow$)	EI ($\downarrow$)	EI^* ($\downarrow$)	Q ($\uparrow$)	Q^* ($\uparrow$)	t [sec]
spanish-school-t2	sHC	11.0	23.1%	0.241	0.379	-0.518	-0.241	0.520	0.131	5.4×10^{-3}
	sRA	10.9	19.9%	0.270	0.348	-0.460	-0.305	0.518	0.156	1.3×10^{-1}
	sSG	7.2	23.9%	0.699	0.495	0.399	-0.009	0.072	0.012	1.3×10^0
	sFLPA	10.3	20.3%	0.268	0.356	-0.464	-0.289	0.518	0.153	1.4×10^{-2}
	sFLPA*	14.1	21.7%	0.384	0.231	-0.232	-0.539	0.405	0.258	2.2×10^{-2}
	sDCSBM	13.6	12.2%	0.785	0.496	0.570	-0.008	0.045	0.008	7.6×10^{-1}
	sDCSBM ²	16.4	12.2%	0.802	0.492	0.604	-0.016	0.041	0.013	2.9×10^{-1}
	sLouvain	7.0	24.4%	0.238	0.384	-0.524	-0.232	0.527	0.143	2.0×10^{-2}
	FLPA	7.4	27.6%	0.250	0.411	-0.501	-0.178	0.509	0.103	4.6×10^{-3}
	DCSBM	10.7	12.5%	0.767	0.493	0.533	-0.015	0.051	0.009	2.0×10^{-1}
	Walktrap	8.0	25.6%	0.684	0.493	0.369	-0.014	0.084	0.018	9.4×10^{-3}
	Louvain	6.8	24.9%	0.246	0.432	-0.509	-0.135	0.511	0.075	2.4×10^{-2}
	Leiden	6.9	24.5%	0.690	0.492	0.380	-0.017	0.081	0.020	2.1×10^{-2}
EGFR-symm	sHC	1.0	100.0%	0.339	0.500	-0.322	0.000	0.000	0.000	1.1×10^{-2}
	sRA	129.6	3.9%	0.385	0.292	-0.231	-0.416	0.270	0.208	1.8×10^{-1}
	sSG	20.9	20.2%	0.585	0.474	0.170	-0.053	0.052	0.030	1.1×10^0
	sFLPA	58.8	10.4%	0.319	0.282	-0.363	-0.435	0.325	0.217	5.5×10^{-3}
	sFLPA*	97.8	8.1%	0.322	0.247	-0.356	-0.507	0.323	0.250	5.5×10^{-3}
	sDCSBM	7.7	23.6%	0.501	0.456	0.003	-0.087	0.103	0.043	6.0×10^{-1}
	sDCSBM ²	8.4	20.9%	0.520	0.467	0.041	-0.065	0.090	0.032	2.8×10^{-1}
	sLouvain	52.1	10.4%	0.301	0.259	-0.399	-0.483	0.345	0.248	3.0×10^{-2}
	FLPA	33.6	22.7%	0.424	0.500	-0.151	-0.000	0.199	0.004	2.0×10^{-3}
	DCSBM	9.4	20.4%	0.525	0.467	0.050	-0.067	0.089	0.030	2.2×10^{-1}
	Walktrap	19.0	20.1%	0.576	0.518	0.153	0.036	0.051	-0.013	3.2×10^{-3}
	Louvain	11.0	15.7%	0.391	0.497	-0.218	-0.006	0.230	0.004	1.2×10^{-2}
	Leiden	11.5	14.5%	0.568	0.504	0.136	0.009	0.061	-0.002	6.6×10^{-3}
spanish-school-t1	sHC	15.0	24.9%	0.278	0.295	-0.445	-0.411	0.451	0.198	2.5×10^{-2}
	sRA	9.1	27.9%	0.269	0.312	-0.461	-0.376	0.445	0.183	4.7×10^{-1}
	sSG	6.0	20.2%	0.602	0.476	0.203	-0.049	0.132	0.028	2.6×10^0
	sFLPA	8.3	27.1%	0.269	0.310	-0.462	-0.380	0.448	0.186	4.5×10^{-2}
	sFLPA*	15.7	22.3%	0.348	0.236	-0.305	-0.529	0.391	0.244	6.2×10^{-2}
	sDCSBM	21.2	10.5%	0.741	0.480	0.482	-0.040	0.069	0.020	1.8×10^0
	sDCSBM ²	29.8	8.1%	0.763	0.480	0.526	-0.040	0.058	0.021	6.3×10^{-1}
	sLouvain	6.1	24.1%	0.262	0.315	-0.476	-0.370	0.463	0.194	7.8×10^{-2}
	FLPA	5.6	32.0%	0.279	0.448	-0.441	-0.105	0.414	0.061	1.4×10^{-2}
	DCSBM	8.8	27.0%	0.598	0.479	0.195	-0.041	0.131	0.029	5.1×10^{-1}
	Walktrap	7.0	25.9%	0.575	0.463	0.149	-0.074	0.145	0.048	2.4×10^{-2}
	Louvain	5.7	28.5%	0.289	0.461	-0.422	-0.077	0.426	0.051	6.7×10^{-2}
	Leiden	5.8	27.4%	0.565	0.469	0.129	-0.063	0.151	0.046	3.0×10^{-2}
us-congress-h114	sHC	2.0	57.0%	0.071	0.058	-0.857	-0.885	0.431	0.443	5.9×10^{-2}
	sRA	27.2	47.7%	0.021	0.024	-0.959	-0.952	0.408	0.424	3.5×10^0
	sSG	2.1	54.3%	0.064	0.052	-0.873	-0.896	0.435	0.447	1.8×10^1
	sFLPA	27.1	47.7%	0.021	0.024	-0.958	-0.952	0.408	0.424	4.0×10^{-1}
	sFLPA*	26.3	50.1%	0.022	0.022	-0.956	-0.956	0.416	0.431	4.3×10^{-1}
	sDCSBM	18.5	19.9%	0.249	0.351	-0.502	-0.298	0.119	0.127	9.9×10^0
	sDCSBM ²	21.2	18.4%	0.264	0.372	-0.472	-0.256	0.104	0.110	4.1×10^0
	sLouvain	2.0	54.3%	0.064	0.052	-0.872	-0.896	0.435	0.447	4.1×10^{-1}
	FLPA	1.0	100.0%	0.646	0.500	0.293	0.000	0.000	0.000	5.3×10^{-2}
	DCSBM	2.2	49.4%	0.196	0.205	-0.607	-0.590	0.287	0.284	1.8×10^1
	Walktrap	1.0	100.0%	0.646	0.500	0.293	0.000	0.000	0.000	2.3×10^{-1}
	Louvain	3.0	39.7%	0.396	0.419	-0.208	-0.162	0.056	0.076	7.3×10^{-1}
	Leiden	3.0	39.8%	0.404	0.427	-0.193	-0.145	0.047	0.066	2.6×10^{-1}

Table C.7: (continued) Statistics of clusterings detected in empirical signed networks using different algorithms. (3/6)

Network	Algorithm	Clusters	Largest	FI ($\downarrow$)	FI^* ($\downarrow$)	EI ($\downarrow$)	EI^* ($\downarrow$)	Q ($\uparrow$)	Q^* ($\uparrow$)	t [sec]
spanish-school-t6	sHC	26.0	14.0%	0.339	0.312	-0.322	-0.375	0.467	0.182	5.5×10^{-2}
	sRA	17.3	18.2%	0.305	0.338	-0.390	-0.323	0.482	0.162	1.0×10^0
	sSG	5.1	25.6%	0.343	0.502	-0.315	0.003	0.348	0.007	3.3×10^0
	sFLPA	16.5	18.4%	0.301	0.342	-0.399	-0.317	0.485	0.159	6.2×10^{-2}
	sFLPA*	27.5	13.4%	0.390	0.250	-0.220	-0.500	0.416	0.241	8.0×10^{-2}
	sDCSBM	25.1	7.4%	0.689	0.484	0.377	-0.033	0.139	0.021	1.8×10^0
	sDCSBM ²	28.1	6.0%	0.695	0.482	0.391	-0.036	0.136	0.022	6.5×10^{-1}
	sLouvain	5.7	25.0%	0.179	0.423	-0.643	-0.155	0.522	0.082	8.4×10^{-2}
	FLPA	10.9	23.1%	0.280	0.477	-0.439	-0.045	0.481	0.038	1.6×10^{-2}
	DCSBM	11.5	25.2%	0.446	0.506	-0.108	0.013	0.295	0.010	4.7×10^{-1}
	Walktrap	4.0	29.2%	0.321	0.486	-0.357	-0.029	0.361	0.022	3.7×10^{-2}
	Louvain	4.0	29.2%	0.163	0.505	-0.674	0.009	0.519	0.004	7.8×10^{-2}
	Leiden	4.0	29.2%	0.321	0.486	-0.357	-0.029	0.361	0.022	4.1×10^{-2}
org-conflict-17	sHC	1.0	100.0%	0.121	0.500	-0.758	0.000	0.000	0.000	5.7×10^{-2}
	sRA	33.9	56.5%	0.212	0.332	-0.577	-0.335	0.262	0.087	8.5×10^{-1}
	sSG	11.2	19.5%	0.775	0.499	0.549	-0.002	0.004	0.003	2.8×10^0
	sFLPA	25.3	60.5%	0.200	0.357	-0.601	-0.285	0.260	0.072	2.0×10^{-2}
	sFLPA*	44.3	35.3%	0.371	0.221	-0.258	-0.557	0.295	0.213	2.7×10^{-2}
	sDCSBM	13.7	19.6%	0.793	0.491	0.585	-0.018	0.017	0.014	1.4×10^0
	sDCSBM ²	19.7	17.6%	0.813	0.485	0.626	-0.029	0.017	0.016	5.2×10^{-1}
	sLouvain	11.8	18.3%	0.366	0.363	-0.269	-0.274	0.416	0.147	4.6×10^{-2}
	FLPA	8.8	68.2%	0.191	0.488	-0.617	-0.024	0.211	-0.028	6.6×10^{-3}
	DCSBM	15.7	22.1%	0.789	0.493	0.577	-0.013	0.017	0.016	3.2×10^{-1}
	Walktrap	24.0	31.6%	0.713	0.552	0.426	0.103	0.012	-0.007	1.2×10^{-2}
	Louvain	8.8	23.5%	0.370	0.494	-0.261	-0.013	0.394	0.025	3.4×10^{-2}
	Leiden	8.8	22.2%	0.751	0.540	0.503	0.080	0.010	-0.011	1.7×10^{-2}
macrophage-symm	sHC	1.0	100.0%	0.335	0.500	-0.329	0.000	0.000	0.000	1.1×10^{-1}
	sRA	270.3	3.4%	0.347	0.264	-0.305	-0.472	0.313	0.236	8.4×10^{-1}
	sSG	32.0	21.3%	0.608	0.487	0.217	-0.025	0.032	0.013	3.0×10^0
	sFLPA	141.3	4.7%	0.296	0.270	-0.407	-0.459	0.361	0.229	8.5×10^{-3}
	sFLPA*	217.6	4.0%	0.301	0.230	-0.398	-0.540	0.357	0.268	9.0×10^{-3}
	sDCSBM	12.6	19.6%	0.554	0.472	0.109	-0.057	0.072	0.028	7.1×10^{-1}
	sDCSBM ²	12.5	16.6%	0.555	0.473	0.111	-0.054	0.077	0.029	3.5×10^{-1}
	sLouvain	76.8	13.8%	0.248	0.223	-0.503	-0.554	0.408	0.290	6.7×10^{-2}
	FLPA	90.4	7.0%	0.439	0.534	-0.123	0.067	0.219	-0.028	3.1×10^{-3}
	DCSBM	20.6	9.5%	0.573	0.474	0.145	-0.052	0.075	0.029	2.6×10^{-1}
	Walktrap	46.0	11.4%	0.564	0.483	0.127	-0.035	0.087	0.020	6.5×10^{-3}
	Louvain	22.9	9.9%	0.384	0.523	-0.232	0.045	0.260	-0.021	2.3×10^{-2}
	Leiden	22.5	9.7%	0.544	0.488	0.089	-0.024	0.102	0.015	1.1×10^{-2}
yeast-symm	sHC	1.0	100.0%	0.204	0.500	-0.593	0.000	0.000	0.000	1.1×10^{-1}
	sRA	148.3	8.1%	0.231	0.146	-0.538	-0.708	0.540	0.347	4.9×10^{-1}
	sSG	28.9	15.3%	0.716	0.472	0.431	-0.056	0.025	0.021	2.8×10^0
	sFLPA	134.3	8.3%	0.205	0.134	-0.591	-0.733	0.563	0.359	4.5×10^{-3}
	sFLPA*	137.1	8.4%	0.205	0.129	-0.590	-0.742	0.563	0.363	4.9×10^{-3}
	sDCSBM	9.0	47.9%	0.602	0.514	0.205	0.027	0.021	0.013	9.1×10^{-1}
	sDCSBM ²	8.4	46.6%	0.602	0.507	0.204	0.014	0.025	0.016	5.2×10^{-1}
	sLouvain	26.7	16.0%	0.107	0.088	-0.785	-0.823	0.674	0.501	3.8×10^{-2}
	FLPA	67.7	8.4%	0.348	0.453	-0.305	-0.094	0.430	0.064	2.4×10^{-3}
	DCSBM	7.2	37.9%	0.631	0.495	0.262	-0.009	0.030	0.024	5.6×10^{-1}
	Walktrap	44.0	16.8%	0.718	0.473	0.435	-0.055	0.028	0.023	7.5×10^{-3}
	Louvain	26.4	11.5%	0.275	0.465	-0.450	-0.070	0.477	0.055	1.6×10^{-2}
	Leiden	26.7	11.2%	0.715	0.480	0.430	-0.040	0.032	0.017	8.8×10^{-3}

Table C.8: (continued) Statistics of clusterings detected in empirical signed networks using different algorithms. (4/6)

Network	Algorithm	Clusters	Largest	FI ($\downarrow$)	FI^* ($\downarrow$)	EI ($\downarrow$)	EI^* ($\downarrow$)	Q ($\uparrow$)	Q^* ($\uparrow$)	t [sec]
ecoli-symm	sHC	1.0	100.0%	0.416	0.500	-0.169	0.000	0.000	0.000	1.1×10^0
	sRA	437.9	14.4%	0.160	0.137	-0.681	-0.726	0.385	0.336	3.6×10^0
	sSG	47.1	34.3%	0.562	0.502	0.124	0.003	0.000	0.002	7.7×10^0
	sFLPA	420.4	15.5%	0.148	0.129	-0.703	-0.742	0.393	0.342	1.5×10^{-2}
	sFLPA*	430.1	15.2%	0.147	0.126	-0.706	-0.748	0.394	0.344	1.7×10^{-2}
	sDCSBM	20.8	38.9%	0.539	0.494	0.078	-0.012	-0.001	0.001	1.9×10^0
	sDCSBM ²	18.0	40.6%	0.538	0.496	0.075	-0.008	-0.003	-0.001	9.6×10^{-1}
	sLouvain	42.2	27.1%	0.122	0.109	-0.756	-0.782	0.461	0.419	1.4×10^{-1}
	FLPA	73.1	19.8%	0.460	0.487	-0.080	-0.025	0.089	0.000	7.4×10^{-3}
	DCSBM	4.9	57.1%	0.504	0.501	0.008	0.002	0.001	0.004	1.9×10^0
	Walktrap	88.0	29.7%	0.577	0.503	0.154	0.007	-0.008	-0.005	3.1×10^{-2}
	Louvain	35.0	14.9%	0.466	0.503	-0.067	0.007	0.091	-0.010	4.6×10^{-2}
	Leiden	36.7	14.1%	0.570	0.499	0.140	-0.003	-0.006	-0.004	2.0×10^{-2}
birdwatch-sg	sFLPA	195.5	59.5%	0.135	0.142	-0.729	-0.716	0.349	0.334	4.7×10^{-1}
	sFLPA*	225.5	56.3%	0.140	0.138	-0.720	-0.723	0.342	0.327	7.1×10^{-1}
	sLouvain	10.9	54.4%	0.475	0.493	-0.051	-0.014	0.125	0.106	1.9×10^0
	FLPA	1.0	100.0%	0.369	0.500	-0.262	0.000	0.000	0.000	4.7×10^{-2}
	Louvain	6.6	28.2%	0.451	0.427	-0.098	-0.146	0.120	0.068	1.8×10^0
	Leiden	8.4	22.8%	0.588	0.501	0.176	0.003	-0.003	-0.003	1.5×10^0
soc-bitcoin-alpha	sFLPA	170.4	78.8%	0.096	0.280	-0.808	-0.441	0.143	0.081	9.6×10^{-2}
	sFLPA*	245.4	64.4%	0.226	0.136	-0.547	-0.729	0.199	0.159	1.3×10^{-1}
	sLouvain	21.1	20.7%	0.362	0.300	-0.277	-0.400	0.436	0.230	6.4×10^{-1}
	FLPA	36.9	91.6%	0.084	0.483	-0.833	-0.033	0.059	0.015	3.2×10^{-2}
	Louvain	17.9	21.9%	0.378	0.467	-0.243	-0.067	0.418	0.074	3.9×10^{-1}
	Leiden	22.9	21.3%	0.788	0.499	0.576	-0.001	0.005	0.000	1.3×10^{-1}
meneame-russia	sFLPA	138.4	67.4%	0.142	0.125	-0.717	-0.751	0.288	0.338	5.1×10^{-1}
	sFLPA*	208.0	63.7%	0.151	0.100	-0.698	-0.801	0.300	0.356	5.9×10^{-1}
	sLouvain	25.3	51.5%	0.190	0.129	-0.620	-0.741	0.317	0.368	1.6×10^0
	FLPA	3.3	99.9%	0.101	0.500	-0.798	0.000	0.000	0.000	1.2×10^{-1}
	Louvain	23.7	47.1%	0.223	0.184	-0.554	-0.631	0.301	0.308	1.3×10^0
	Leiden	23.3	47.0%	0.609	0.495	0.218	-0.010	0.000	0.001	5.0×10^{-1}
soc-bitcoin-otc	sFLPA	441.8	75.5%	0.095	0.196	-0.810	-0.608	0.135	0.088	1.6×10^{-1}
	sFLPA*	557.1	56.3%	0.204	0.132	-0.592	-0.736	0.231	0.170	2.1×10^{-1}
	sLouvain	20.0	20.2%	0.335	0.249	-0.329	-0.503	0.442	0.294	1.2×10^0
	FLPA	52.8	89.2%	0.120	0.455	-0.761	-0.089	0.063	-0.034	5.2×10^{-2}
	Louvain	19.4	20.0%	0.388	0.486	-0.225	-0.028	0.388	0.027	5.7×10^{-1}
	Leiden	22.4	19.4%	0.675	0.449	0.351	-0.102	0.093	0.034	2.2×10^{-1}
chess	sFLPA	1415.2	23.3%	0.338	0.313	-0.324	-0.374	0.187	0.160	7.3×10^{-1}
	sFLPA*	1598.9	4.8%	0.385	0.333	-0.230	-0.335	0.192	0.166	6.8×10^{-1}
	sLouvain	228.3	15.9%	0.347	0.307	-0.305	-0.386	0.224	0.195	2.8×10^0
	FLPA	222.5	87.9%	0.420	0.498	-0.161	-0.003	0.005	-0.004	9.2×10^{-2}
	Louvain	118.4	15.3%	0.479	0.506	-0.042	0.011	0.082	-0.008	8.4×10^{-1}
	Leiden	115.7	14.5%	0.479	0.506	-0.042	0.012	0.085	-0.008	3.0×10^{-1}
elec	sFLPA	1164.4	62.4%	0.182	0.276	-0.636	-0.448	0.114	0.072	8.7×10^{-1}
	sFLPA*	1211.6	40.9%	0.274	0.216	-0.451	-0.567	0.183	0.137	1.4×10^0
	sLouvain	41.6	31.5%	0.338	0.308	-0.324	-0.385	0.318	0.197	2.6×10^0
	FLPA	31.1	91.9%	0.229	0.498	-0.542	-0.004	0.031	0.001	1.9×10^{-1}
	Louvain	30.9	30.3%	0.394	0.487	-0.213	-0.027	0.251	0.014	1.5×10^0
	Leiden	30.3	29.0%	0.397	0.487	-0.205	-0.027	0.253	0.014	5.0×10^{-1}

Table C.9: (continued) Statistics of clusterings detected in empirical signed networks using different algorithms. (5/6)

Network	Algorithm	Clusters	Largest	FI ($\downarrow$)	FI^* ($\downarrow$)	EI ($\downarrow$)	EI^* ($\downarrow$)	Q ($\uparrow$)	Q^* ($\uparrow$)	t [sec]
meneame-politics	sFLPA	281.2	94.6%	0.093	0.393	-0.814	-0.214	0.002	0.009	2.3×10^0
	sFLPA*	380.3	68.4%	0.193	0.143	-0.613	-0.714	0.181	0.261	3.1×10^0
	sLouvain	19.3	55.1%	0.258	0.190	-0.484	-0.620	0.218	0.291	6.5×10^0
	FLPA	2.5	100.0%	0.103	0.500	-0.795	0.000	0.000	0.000	3.8×10^{-1}
	Louvain	15.2	43.3%	0.348	0.286	-0.304	-0.429	0.190	0.196	5.9×10^0
	Leiden	19.1	44.8%	0.599	0.504	0.198	0.008	0.000	-0.000	1.9×10^0
compling-authors	sFLPA	227.9	92.8%	0.083	0.461	-0.835	-0.077	0.009	0.003	8.3×10^{-1}
	sFLPA*	290.3	50.3%	0.462	0.303	-0.075	-0.394	0.158	0.126	3.0×10^0
	sLouvain	33.4	19.3%	0.452	0.428	-0.096	-0.144	0.327	0.078	1.9×10^0
	FLPA	49.1	95.7%	0.091	0.500	-0.819	-0.000	0.012	0.002	2.1×10^{-1}
	Louvain	27.9	22.0%	0.449	0.502	-0.102	0.004	0.326	0.001	2.4×10^0
	Leiden	28.9	21.8%	0.800	0.498	0.600	-0.004	0.002	0.002	8.3×10^{-1}
compling-papers	sFLPA	1 044.3	18.9%	0.316	0.298	-0.369	-0.404	0.509	0.194	2.3×10^{-1}
	sFLPA*	1 214.0	6.9%	0.433	0.238	-0.133	-0.524	0.460	0.258	2.4×10^{-1}
	sLouvain	336.5	10.9%	0.262	0.306	-0.476	-0.387	0.592	0.194	9.9×10^{-1}
	FLPA	698.1	22.7%	0.322	0.501	-0.355	0.002	0.479	-0.001	9.4×10^{-2}
	Louvain	166.6	12.2%	0.261	0.505	-0.479	0.009	0.576	-0.003	6.7×10^{-1}
	Leiden	164.1	11.4%	0.840	0.503	0.680	0.005	0.020	-0.003	2.5×10^{-1}
wiki-rfa	sFLPA	1 642.2	45.9%	0.236	0.302	-0.529	-0.396	0.243	0.119	1.9×10^0
	sFLPA*	1 716.6	30.1%	0.323	0.249	-0.354	-0.503	0.264	0.171	2.9×10^0
	sLouvain	6.4	31.3%	0.316	0.317	-0.367	-0.366	0.349	0.193	4.8×10^0
	FLPA	3.7	76.4%	0.256	0.496	-0.489	-0.008	0.116	0.005	4.8×10^{-1}
	Louvain	5.0	26.4%	0.370	0.484	-0.260	-0.032	0.288	0.018	2.7×10^0
	Leiden	5.1	24.7%	0.370	0.484	-0.260	-0.032	0.290	0.018	9.4×10^{-1}
soc-reddit-body	sFLPA	2 359.2	43.8%	0.234	0.472	-0.532	-0.056	0.507	0.173	1.1×10^0
	sFLPA*	3 558.5	2.9%	0.496	0.288	-0.007	-0.424	0.425	0.214	1.0×10^0
	sLouvain	583.4	21.4%	0.276	0.456	-0.448	-0.087	0.581	0.193	1.0×10^1
	FLPA	1 455.7	50.8%	0.222	0.514	-0.557	0.029	0.489	0.148	3.7×10^{-1}
	Louvain	552.2	17.5%	0.288	0.465	-0.424	-0.070	0.575	0.135	4.8×10^0
	Leiden	572.8	16.7%	0.274	0.465	-0.452	-0.069	0.595	0.137	1.4×10^0
soc-reddit-title	sFLPA	3 007.7	69.0%	0.180	0.491	-0.639	-0.018	0.221	0.090	1.9×10^0
	sFLPA*	4 230.7	31.3%	0.504	0.309	0.008	-0.382	0.271	0.164	3.8×10^0
	sLouvain	841.4	16.9%	0.399	0.463	-0.203	-0.075	0.407	0.156	3.2×10^1
	FLPA	1 764.6	75.5%	0.170	0.513	-0.660	0.026	0.190	0.069	5.4×10^{-1}
	Louvain	803.8	16.8%	0.405	0.454	-0.190	-0.092	0.405	0.086	1.1×10^1
	Leiden	841.0	14.0%	0.404	0.446	-0.193	-0.107	0.421	0.085	2.7×10^0
soc-Slashdot090221	sFLPA	10 407.2	74.9%	0.143	0.211	-0.713	-0.578	0.072	0.058	4.0×10^0
	sFLPA*	11 664.9	63.5%	0.196	0.158	-0.607	-0.684	0.111	0.095	7.1×10^0
	sLouvain	460.6	23.4%	0.388	0.289	-0.224	-0.423	0.318	0.252	5.9×10^1
	FLPA	1 445.9	94.0%	0.230	0.501	-0.539	0.003	0.019	0.011	8.5×10^{-1}
	Louvain	439.3	25.3%	0.468	0.486	-0.063	-0.029	0.223	0.044	1.5×10^1
	Leiden	452.7	24.7%	0.686	0.497	0.371	-0.005	0.003	0.001	4.5×10^0
wikiconflict	sFLPA	70 566.1	34.5%	0.076	0.075	-0.849	-0.850	0.108	0.098	1.9×10^1
	sFLPA*	70 615.1	33.8%	0.082	0.081	-0.835	-0.838	0.112	0.105	1.6×10^1
	sLouvain	3 598.6	45.4%	0.279	0.297	-0.442	-0.407	0.238	0.243	2.3×10^2
	FLPA	2 497.3	93.3%	0.567	0.502	0.134	0.004	-0.012	-0.009	3.5×10^0
	Louvain	1 820.9	7.7%	0.698	0.690	0.395	0.381	-0.265	-0.198	3.7×10^1
	Leiden	1 836.6	14.2%	0.670	0.658	0.340	0.317	-0.237	-0.171	1.4×10^1

Table C.10: (continued) Statistics of clusterings detected in empirical signed networks using different algorithms. (6/6)

Network	Algorithm	Clusters	Largest	FI ($\downarrow$)	FI^* ($\downarrow$)	EI ($\downarrow$)	EI^* ($\downarrow$)	Q ($\uparrow$)	Q^* ($\uparrow$)	t [sec]
soc-epinions	sFLPA	28 277.1	56.9%	0.088	0.161	-0.823	-0.678	0.211	0.155	5.3×10^0
	sFLPA*	29 737.6	48.4%	0.140	0.113	-0.719	-0.773	0.237	0.177	6.8×10^0
	sLouvain	7 422.1	21.6%	0.300	0.236	-0.400	-0.527	0.446	0.313	8.3×10^1
	FLPA	9 182.0	77.2%	0.162	0.492	-0.676	-0.016	0.162	0.078	1.3×10^0
	Louvain	6 878.1	21.8%	0.355	0.474	-0.291	-0.051	0.380	0.073	2.2×10^1
	Leiden	6 921.3	20.0%	0.760	0.498	0.519	-0.003	0.009	0.003	6.1×10^0
wiki-users	sFLPA	17 008.8	59.1%	0.101	0.179	-0.798	-0.642	0.156	0.088	5.6×10^0
	sFLPA*	18 438.5	40.4%	0.213	0.148	-0.575	-0.703	0.263	0.161	8.9×10^0
	sLouvain	614.7	18.0%	0.342	0.246	-0.316	-0.508	0.490	0.274	8.7×10^1
	FLPA	1 493.5	91.9%	0.138	0.490	-0.723	-0.020	0.059	0.012	1.7×10^0
	Louvain	360.5	15.2%	0.378	0.418	-0.244	-0.165	0.447	0.095	2.5×10^1
	Leiden	357.2	17.1%	0.374	0.433	-0.251	-0.134	0.456	0.091	5.4×10^0